

Mensah Philemon Edem Yao

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Email | LinkedIn | Portfolio

Education

University of Ghana, Legon

Jan 2024 – Present

BSc Computer Engineering

- First Class Academic Standing

- Key Courses: C++, Calculus, Linear Algebra, Database Management, Software Engineering, Data Structures and Algorithms

Work Experience

AI Intern, Remotown

Sep 2025 – Present

- Designed and deployed job and company canonicalization pipelines using pretrained transformer models, improving data quality and reducing duplicate entities across datasets.

- Cleaned, normalized, and preprocessed large-scale noisy job and company data, integrating pipelines directly into automated web scraping systems for real-time ingestion.

- Implemented a vector-based semantic retrieval layer using Qdrant to support Retrieval-Augmented Generation (RAG) on normalized data.

- Migrated backend infrastructure from MongoDB to PostgreSQL, redesigning schemas and optimizing SQL queries to reduce request latency from over 10 minutes to under 1 second.

- Implemented unit and integration tests to ensure backend services and data pipelines remained deterministic and production-ready.

- Built CI/CD pipelines with GitHub Actions and containerized services using Docker, reducing deployment failures by over 70% and ensuring environment consistency across production stages.

Projects

GSBE UG Website for Biomedical Engineering Students

Jun 2025 – Present

Lead Backend Developer — Django, Supabase, PostgreSQL

- Led backend development of a production web platform, designing scalable database schemas and authentication workflows using Django and PostgreSQL.

- Implemented role-based access control and secure authentication, supporting 40+ active users within the department.

- Deployed and maintained the platform in production, ensuring reliability and maintainable backend architecture.

Educational AI-Integrated Web Platform

Apr 2025 – Jun 2025

Django, React, PostgreSQL, Retrieval-Augmented Generation (RAG)

- Developed an AI-powered educational platform integrating RAG for contextual question answering over academic content.

- Built document ingestion and embedding pipelines to enable semantic search and accurate AI-driven responses.

- Integrated backend AI services with a React frontend for real-time, context-aware interactions.

Skills

- Programming: Python, SQL (PostgreSQL)
- Backend & APIs: Django, FastAPI, REST APIs, GraphQL
- AI / ML: Retrieval-Augmented Generation (RAG), Transformer Models, Semantic Search
- Vector Databases: Qdrant
- Data & Analytics: Pandas, NumPy, Matplotlib, Data Preprocessing
- DevOps & Tools: Docker, Git, GitHub, CI/CD (GitHub Actions)

Activities / Memberships

- Member, IEEE Student Chapter, University of Ghana (2025 – Present)